SRIRANJAN KAPILAN

Full Stack Software Engineer

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Summary

Full Stack Software Engineer with 3.5+ years of experience building and operating production applications end to end: React, Next.js and TypeScript frontends, Node.js/Express.js and Java/Spring Boot backend services, and PostgreSQL/MySQL data models with Flyway-managed migrations. Experienced with Redis caching, Apache Kafka for asynchronous processing and microservice architectures, with exposure to fintech and payments, from fintech REST APIs to a production POS platform with multi-payment sales and cash-flow tracking. Build LLM-powered features using function calling with access-scoped, read-only tools, and work with AI-assisted development tools (Claude, Gemini, OpenCode). Own delivery through to production with Docker, Jenkins CI/CD and AWS, backed by monitoring with Sentry, Prometheus, Grafana and Checkmk.

Education

Uva Wellassa University of Sri Lanka *2019 - 2023*
B.Sc. (Hons) in Computer Science and Technology

Technical Skills

Frontend: React.js, Next.js, TypeScript, Tailwind CSS, HTML, Bootstrap.

Backend: Node.js, Express.js, Java, Spring Boot, Spring Security (JWT), Python, Flask, REST APIs, Microservices (Feign Client, WebClient, API Gateway), Apache Kafka.

Databases & Caching: PostgreSQL, MySQL, MongoDB, Redis, Supabase.

AI-Assisted Development: Claude, Gemini, OpenCode.

DevOps & Cloud: Docker, Docker Compose, Git, Jenkins, Nexus, SonarQube, AWS (EC2, RDS, S3, IAM, VPC), Ansible, Linux, Nginx, HAProxy, Cloudflare, Proxmox Virtualisation (HA cluster, VMs, LXC), Proxmox Backup Server, Hetzner Cloud, pfSense, Kata Containers (microVMs), Bash (shell scripting)

Observability & Security: Sentry, Prometheus, Grafana, Checkmk, Matomo, Wazuh SIEM, CrowdSec, PatchMon, Headscale/Tailscale (zero-trust mesh VPN).

Testing: JUnit 5, Mockito, Selenium, Postman.

Working knowledge: Kubernetes.

Operating Systems: Linux (Debian, Ubuntu)

Professional Experience

Software Engineer, HomeIT System *Jan 2025 - Present*

- Developed secure Spring Boot APIs for end-to-end encrypted file transfer to Hetzner S3 using RSA public/private-key encryption, backed by PostgreSQL with Flyway-managed migrations and Redis caching, with Apache Kafka handling asynchronous downstream processing.
- Shipped a production Next.js client website with optimised images and structured SEO metadata, achieving Lighthouse scores of 85+ (SEO) and 80+ (performance), with Sentry integrated for real-time error tracking and performance monitoring; rolled out self-hosted Matomo across client sites for first-party analytics on traffic, page engagement, scroll depth and click paths.
- Contributed to securing a medical practice's infrastructure: a Node.js ops dashboard for security alerts and patch status, isolated VLANs, default-deny firewall rules, and Ansible-automated deployment of Wazuh SIEM behind Traefik and Kata Containers microVMs.
- Engineered a Docker-based CI/CD pipeline publishing images to Nexus and deploying through Dokploy with Docker Compose, using Jenkins agents to automate build and test cycles and cutting deployment time from 30–40 minutes to 5–10 minutes.
- Added HTTP health checks that push Google Chat alerts within 5 minutes of any client site returning a non-200 response, consistently catching outages before clients reported them, while hardening and monitoring 10+ production hosts with Checkmk, Wazuh SIEM, CrowdSec and PatchMon.
- Automated AWS infrastructure with Ansible, provisioning EC2 instances inside a VPC, RDS databases and S3 storage with IAM least-privilege access, giving consistent, repeatable deployments without manual console setup.
- Automated provisioning of 25+ VMs and LXC containers across Proxmox and Hetzner Cloud using Ansible, cutting server setup time from roughly 2 hours to 15 minutes and running weekly Ansible-driven patch cycles with no manual overhead.
- Built a two-node Proxmox HA cluster on Hetzner dedicated hardware with encrypted, incremental off-site backups via Proxmox Backup Server to NAS storage, reducing infrastructure spend by 65% versus equivalent managed cloud while holding 99% uptime.

- Deployed self-hosted open-source business apps as Docker containers via Dokploy, each on its own subdomain behind HAProxy with TLS termination, and secured network access with pfSense, Cloudflare, Nginx and AdGuard, restricting internal tools to Headscale/Tailscale with per-user ACLs.

Associate Software Engineer, CodeLantic Pvt Ltd *Oct 2023 - Jan 2025*

- Decomposed a monolithic application into independent, separately deployable microservices, implementing inter-service communication with Feign Client, WebClient and API Gateway, and validated each extracted service through API-level and full-system testing to confirm feature parity before cutover.
- Built and maintained microservices across 3 projects using Java 17 and Spring Boot, including a retail operations platform (finance, payroll, compliance) and a real estate transactions platform (2FA, property search, KPI dashboards).
- Translated ER diagrams into relational MySQL schemas managed through Flyway migrations, plus complementary MongoDB models using embedded and referenced document design.
- Delivered secure APIs with Spring Security covering PDF and Excel report generation, bulk ZIP downloads and AWS S3 upload/download for multimedia content management.

Software Engineer – Trainee, CodeLantic Pvt Ltd *Mar 2023 - Sep 2023*

- Developed REST APIs with Spring Boot and Java 8 for a fintech application, implementing backend validation with Hibernate Validator and verifying endpoints through Postman.
- Wrote JUnit 5 and Mockito unit tests achieving 80%+ code coverage, resolving code duplication and quality issues to pass SonarQube quality gates.
- Created Selenium automation scripts for full UI flow testing, validating filter functionality against database results to confirm data accuracy.

Projects

Point-of-Sale & E-Commerce Platform | *React, TypeScript, Tailwind CSS, TanStack Query, Express, PostgreSQL (Supabase), Nginx, Cloudflare*

- Built a production POS and e-commerce platform for a retail electronics business, now in daily use: a role-based admin back office for sales, inventory, purchasing, expenses and reporting, plus a public customer storefront, with Supabase auth and RBAC.
- Engineered multi-payment sales and cash-flow tracking with exportable PDF/Excel reporting; deployed to a self-managed VPS behind Nginx and Cloudflare.

Team Weekly Reporting & Review Platform | *Java 21, Spring Boot 4, Spring Security (JWT), MySQL, Flyway, React 19, TypeScript, Tailwind CSS, Google Gemini API, Docker, Jenkins, SonarQube*

- Built a full-stack weekly reporting platform that replaced ad-hoc email and document status updates with a structured submit–review–approve workflow covering tasks, blockers, achievements and hours, with project and user administration and role-based access control.
- Built an AI chat assistant using LLM function calling, whose read-only tools inherit the application's access rules, alongside a manager analytics dashboard.
- Engineered immutable report versioning so corrections never overwrite history, keeping the original submission and reviewer comment as a permanent audit trail; containerised with Docker and published to a private Docker Hub registry via a Jenkins CI/CD pipeline, with SonarQube static analysis passing the quality gate.

Event Management Platform | *React, TypeScript, Tailwind CSS, Node.js, Express, PostgreSQL (Supabase), JWT, QR Code, Nginx, PM2*

- Built a full-stack platform for multi-day event registration, attendance, logistics and staff management, with a validated bulk spreadsheet import pipeline.
- Implemented idempotent QR-code check-in (duplicate-scan prevention, partial family attendance), enforcing meal and certificate uniqueness at the database level and surfacing an "already claimed" indicator during scans; deployed on a self-managed Linux server (Nginx, PM2).

Cleaning Services Booking Website | *Next.js, React, TypeScript, Tailwind CSS, Jenkins, SonarQube*

- Built a conversion-focused marketing and booking site with a multi-step booking wizard and a live pricing engine adjusting in real time across cleaning types, room counts, add-ons and recurring-visit discounts.
- Implemented serverless lead capture (spreadsheet CRM with WhatsApp handoff) and generated SEO-optimised landing pages per service area from a single template; delivered a responsive, animated UI with light/dark theme switch, deployed as a static site through a Jenkins CI/CD pipeline with SonarQube analysis on every build.